

Task 4 — Advanced Security Monitoring & Incident Response

Screenshots

1. Sample logs generated to demonstrate brute force login

```
sample_logs.txt x README.md x
sample_logs.txt
1 2026-09-01 10:00:01 LOGIN_FAILED user=admin ip=192.168.1.50
2 2026-09-01 10:00:03 LOGIN_FAILED user=admin ip=192.168.1.50
3 2026-09-01 10:00:05 LOGIN_FAILED user=admin ip=192.168.1.50
4 2026-09-01 10:00:07 LOGIN_FAILED user=admin ip=192.168.1.50
5 2026-09-01 10:00:09 LOGIN_FAILED user=admin ip=192.168.1.50
6 2026-09-01 10:00:15 LOGIN_SUCCESS user=admin ip=192.168.1.50
7 2026-09-01 10:05:00 LOGIN_SUCCESS user=jsmith ip=192.168.1.20
8 2026-09-01 11:00:00 FILE_ACCESS user=jsmith file=payroll.xlsx ip=192.168.1.20
9 2026-09-01 12:30:00 LOGIN_FAILED user=root ip=203.0.113.99
10 2026-09-01 12:30:02 LOGIN_FAILED user=root ip=203.0.113.99
11 2026-09-01 12:30:04 LOGIN_FAILED user=root ip=203.0.113.99
12 2026-09-01 03:15:00 FILE_ACCESS user=jsmith file=payroll.xlsx ip=203.0.113.99
13
```

2. Terminal output

```
PS C:\Users\USER\OneDrive\Desktop\Internship\GIST_Internship\GIST_Internship\TS3> python log_monitor.py

===== SECURITY MONITORING REPORT =====
Total log entries analyzed: 12
Total alerts generated: 2

[HIGH] Possible brute-force attack: user='admin' from ip=192.168.1.50 had 5 failed login attempts.
[MEDIUM] User 'jsmith' accessed sensitive files from multiple IP addresses: 203.0.113.99, 192.168.1.20. Possible compromised account.

===== RECOMMENDATIONS =====
- Enable account lockout after 3-5 failed login attempts.
- Enable multi-factor authentication (MFA) for all accounts.
```